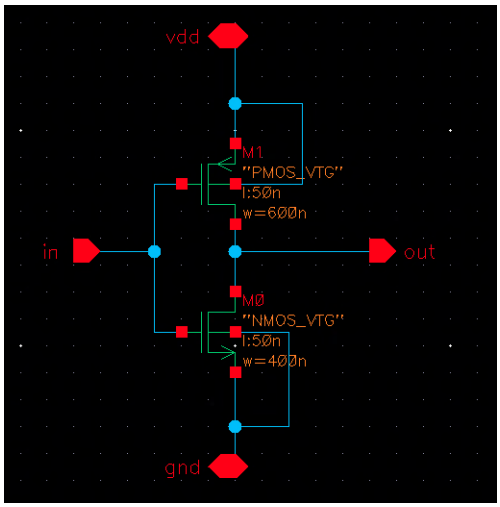
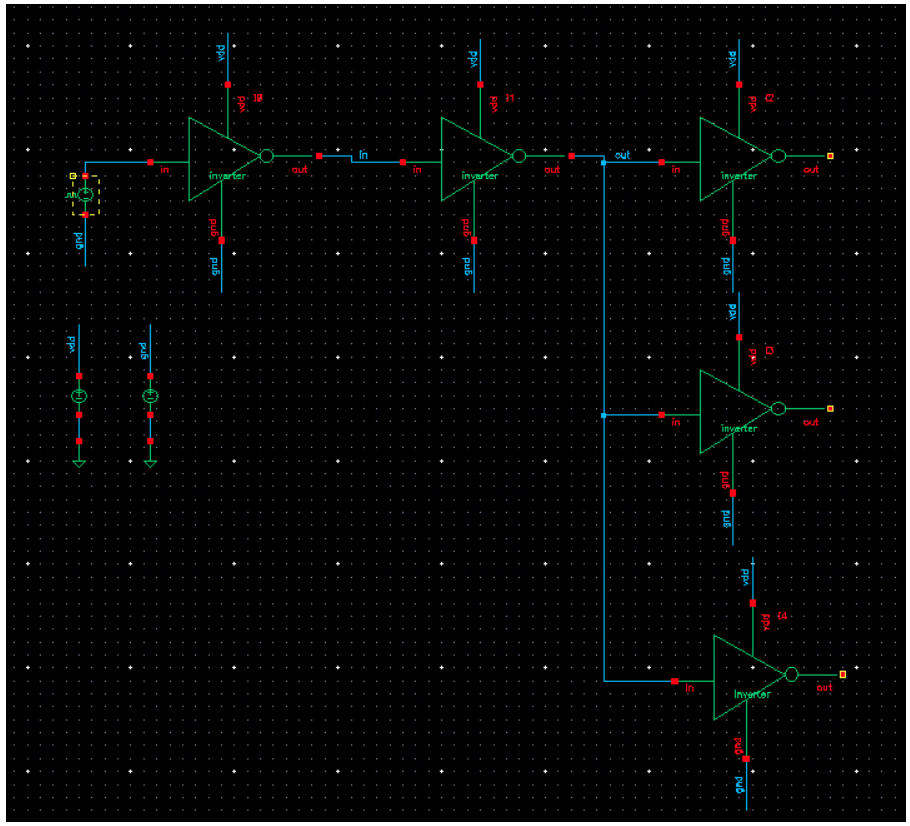
1

Snapshot of your schematics, both inverter and testbench. **Answer:**

Figure 1 shows the inverter schematic (Fig. 1a) and the corresponding testbench (Fig. 1b) used for transient simulations.



(a) Inverter schematic.



(b) Inverter testbench schematic.

Figure 1: Schematics used in this lab: (a) inverter cell and (b) testbench.

2

Snapshot of your layout of single inverter (use ruler shortcut key "k" to show the size of the circuit (from top to bottom, from left to right).

Answer:

Figure 2 shows the finalized inverter layout with ruler measurements (left-to-right $\approx 0.365\,\mu$m; top-to-bottom $\approx 1.8625\,\mu$m indicated in the ruler view).

3

Snapshot of DRC results showing you are DRC clean.

Answer:

As shown in Fig. 3, the layout passes all DRC checks (DRC clean).

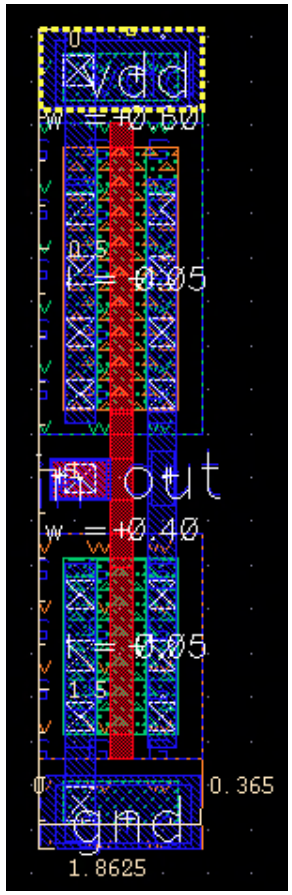
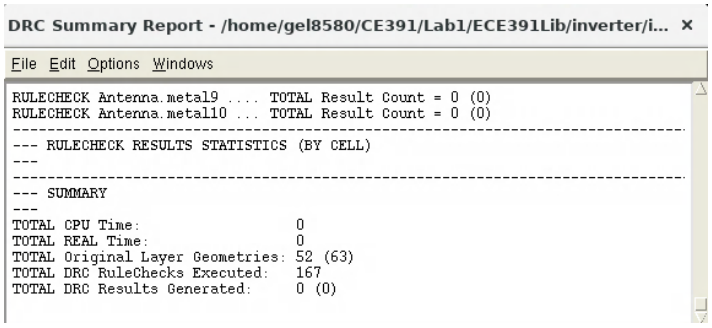


Figure 2: Single inverter layout with ruler measurements.



DRC Summary Report - /home/gel8580/CE391/Lab1/ECE391Lib/inverter/i...

File Edit Options Windows

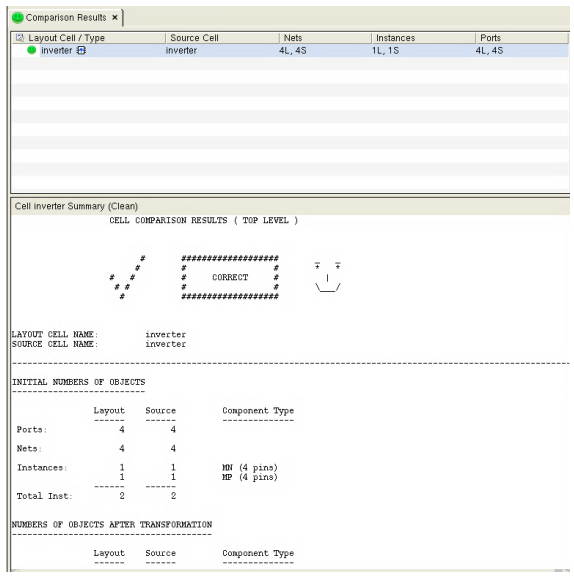
```
RULECHECK Antenna.metal9 .... TOTAL Result Count = 0 (0)
RULECHECK Antenna.metal10 ... TOTAL Result Count = 0 (0)
--------------------------------------------------------------------------------
--- RULECHECK RESULTS STATISTICS (BY CELL)
---
--------------------------------------------------------------------------------
--- SUMMARY
---
TOTAL CPU Time:                  0
TOTAL REAL Time:                 0
TOTAL Original Layer Geometries: 52 (63)
TOTAL DRC RuleChecks Executed:   167
TOTAL DRC Results Generated:     0 (0)
```

Figure 3: DRC report showing the layout is DRC clean.

4

Snapshot of LVS results showing you are LVS clean.
Answer:

Figure 4 confirms the extracted layout netlist matches the schematic netlist (LVS clean).



Comparison Results

Layout Cell / Type	Source Cell	Nets	Instances	Ports
inverter	inverter	4L, 4S	1L, 1S	4L, 4S

Cell inverter Summary (Clean)

```
                    CELL COMPARISON RESULTS ( TOP LEVEL )

                        #       ###################
                       #        #                 #
                    #   #       #     CORRECT     #
                     # #        #                 #
                      #         ###################

LAYOUT CELL NAME:          inverter
SOURCE CELL NAME:          inverter
--------------------------------------------------------------------------------

INITIAL NUMBERS OF OBJECTS
--------------------------

                  Layout    Source         Component Type
                  ------    ------         --------------
 Ports:              4         4

 Nets:               4         4

 Instances:          1         1          MN (4 pins)
                     1         1          MP (4 pins)
                  ------    ------
 Total Inst:         2         2

NUMBERS OF OBJECTS AFTER TRANSFORMATION
---------------------------------------

                  Layout    Source         Component Type
                  ------    ------         --------------
```

Figure 4: LVS result showing layout versus schematic matches.

5

Snapshot of Layout extraction summary.
Answer:

The post-layout RC extraction summary is shown in Fig. 5, including the resistor counts and total/coupling capacitances for each net.

inverter

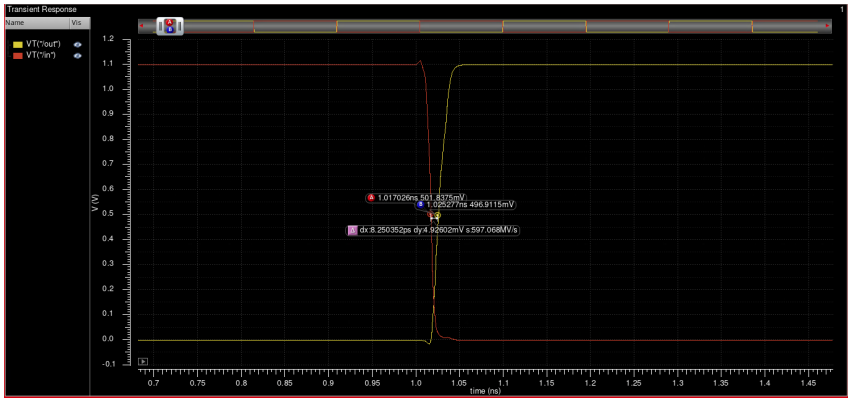
No.	Layout Net	Source Net	R Count	C Total (F)	CC Total (F)	C+CC Total (F)
1	in	in	8	1.20863E-16	7.81680E-17	1.99031E-16
2	gnd	gnd	4	5.18507E-17	5.84889E-17	1.10340E-16
3	vdd	vdd	4	6.50298E-17	6.84639E-17	1.33494E-16
4	out	out	2	5.75663E-17	1.21824E-16	1.79390E-16

Figure 5: Post-layout RC extraction summary (per-net R count and capacitance breakdown).

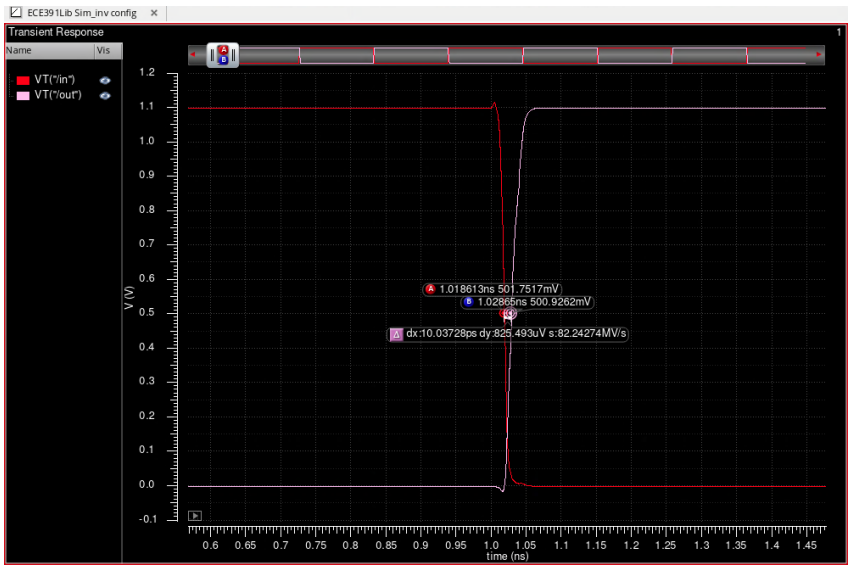
6

Both schematic simulation and layout extraction simulation waveforms of the inverter testbench showing input and output transitions. (Use shortcut key “a” and “b” in the waveform window to measure the delay from the input signal to output signal at 50% transition points.
Answer:

Figure 6 shows the input/output transitions for both schematic and post-layout simulations. Using the 50% crossing points (markers A and B), the measured delay is approximately 8.25 ps for the schematic case (Fig. 6a) and 10.04 ps for the post-layout case (Fig. 6b).



(a) Schematic (front-end) transient waveform and delay measurement at 50% points.



(b) Post-layout (back-end) transient waveform and delay measurement at 50% points.

Figure 6: Inverter transient waveforms (V_{in} and V_{out}) and propagation delay measured at 50% transition points.

7

A simulation result table comparing (1) delay time (both rising and falling), (2) transition time from 20% to 80% for both rising and falling, (3) energy during rise and fall transition of the inverter from both schematic and layout extracted simulation.

Answer:

Let $t_x(\text{sig}, \uparrow)$ denote the time when `VT("/sig")` crosses 0.55 V on the rising edge, and $t_x(\text{sig}, \downarrow)$

denote the crossing time on the falling edge. The metrics in Fig. 7 are defined as:

$$t_{\text{delay_rise}} = t_x(\text{out}, \uparrow) - t_x(\text{in}, \downarrow), \tag{1}$$

$$t_{\text{delay_fall}} = t_x(\text{out}, \downarrow) - t_x(\text{in}, \uparrow), \tag{2}$$

$$E_{\text{rise+fall}} = V_{DD} \int_{20\,\text{ns}}^{30\,\text{ns}} I_{VDD}(t)\, dt. \tag{3}$$

The measured delay and energy results are summarized from ADE outputs in Fig. 7 (front-end: Fig. 7a, back-end: Fig. 7b).

Outputs

	Name/Signal/Expr	Value	Plot	Save
1	in		☐	☑
2	I1/vdd		☐	☑
3	delay_rise	9.534p	☑	☐
4	delay_fall	9.065p	☑	☐
5	energy_rise_fall	7.818f	☑	☐

(a) Schematic (front-end) measured results.

	Name/Signal/Expr	Value	Plot	Save	Save Options
1	in		☐	☑	allv
2	I1/vdd		☐	☑	yes
3	delay_rise	11.46p	☑	☐	
4	delay_fall	11.01p	☑	☐	
5	energy_rise_fall	9.358f	☑	☐	

(b) Post-layout (back-end) measured results.

Figure 7: Simulation outputs reported by ADE for schematic and post-layout extracted simulations.